A DIRECT APPROACH TO TESTING OPTION THEORIES?

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Introduction and Motivation

- How do we use option prices to predict future outcomes?
- Compute Risk-Neutral Density (RND) (Figlewski, 2018)
  - Use market option prices over a range of strike prices, for a given expiry date, to infer the implied probability distribution of the price of the underlying at expiry.
  - RND applies to a particular date in the future; there is ONLY ONE realisation to compare it with.
- Compare Implied Volatility (IV) with future realised volatility (RV) (Poon and Granger, 2003)
  - IV > RV Variance Risk Premium (VRP)
  - But computation of IV requires model, e.g. Black-Scholes
  - And how do we measure RV, e.g. using returns over which time period?

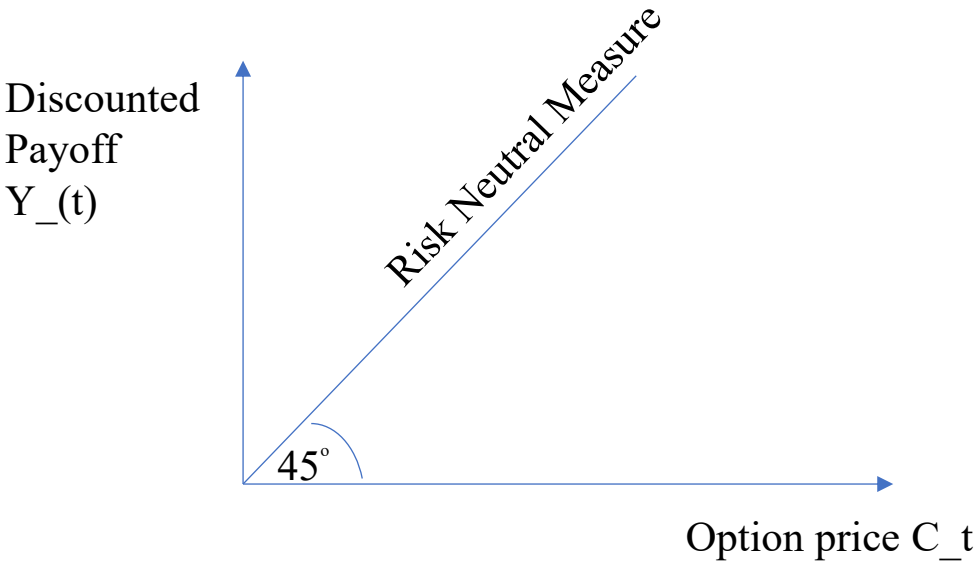
Introduction and Motivation

- Why don't we just compare market option prices with final payoffs? After all, under risk-neutral measure:
  - Option Price at Settlement Date = Discounted Expected Value of Payoff at Expiry
- We can make this comparison using data from all settlement dates, all expiry dates, and all strike prices.

Framework

- Look at (European) option prices on settlement date (t)
  - E.g. price of European Call Option C_t
- Look at payoffs from same option on expiry date (T):
  - Payoff $= \mathrm{Max}(S_T - K, 0)$
- Discount payoff using risk-free rate (r):
  - Discounted Payoff: $Y_{(t)} = \exp[-r(T-t)]\,\mathrm{Max}(S_T - K, 0)$
- Under Risk-Neutral measure: $E(Y_{(t)}) = C_t$
- Leads to regression: $Y_{(t)} = \beta_0 + \beta_1 C_t + \varepsilon_t$

Framework

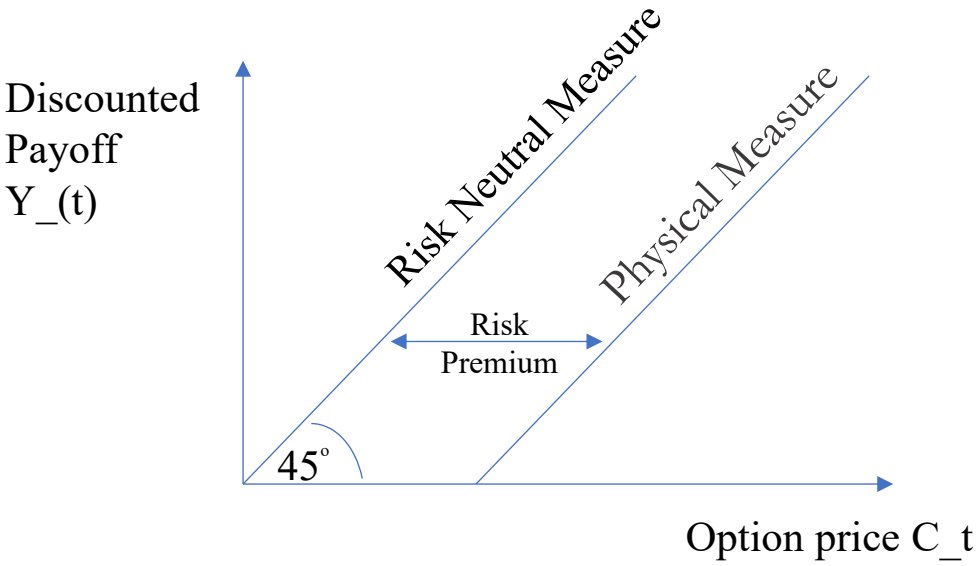


$$Y_{(t)} = \beta_0 + \beta_1 C_t + \varepsilon_t$$

Under risk-neutral measure:

$$\beta_0 = 0; \beta_1 = 1$$

Framework



$$Y_{(t)} = \beta_0 + \beta_1 C_t + \varepsilon_t$$

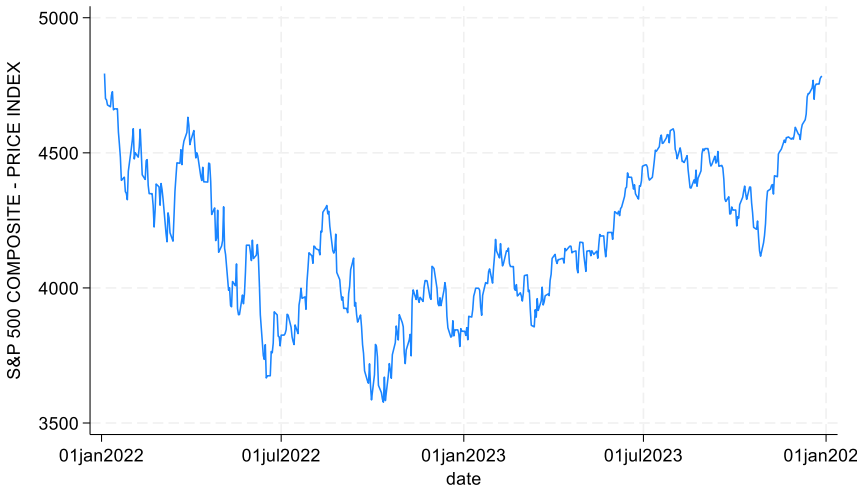
Under Physical measure:

$\beta_0 < 0; \beta_1 = 1$

$-\beta_0$ represents Risk Premium

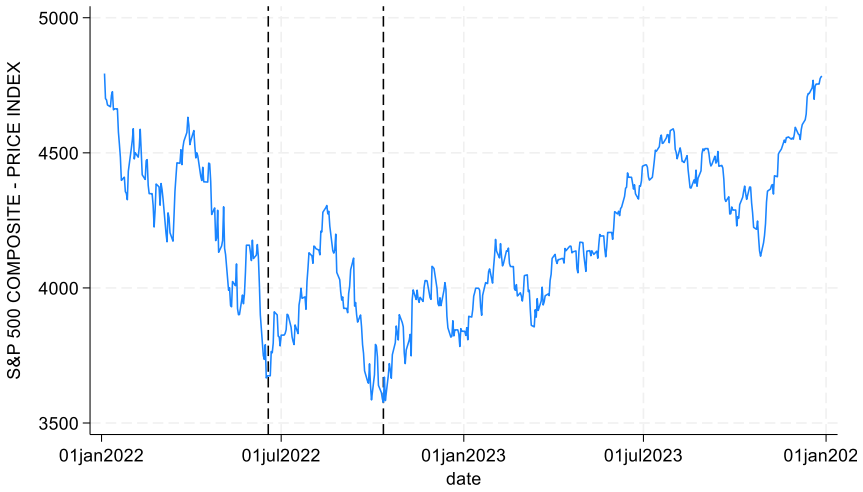
DATA

- All European Calls and European Puts written on S&P500 Index
  - Settlement between 3 Jan 2022 and 29 December 2023

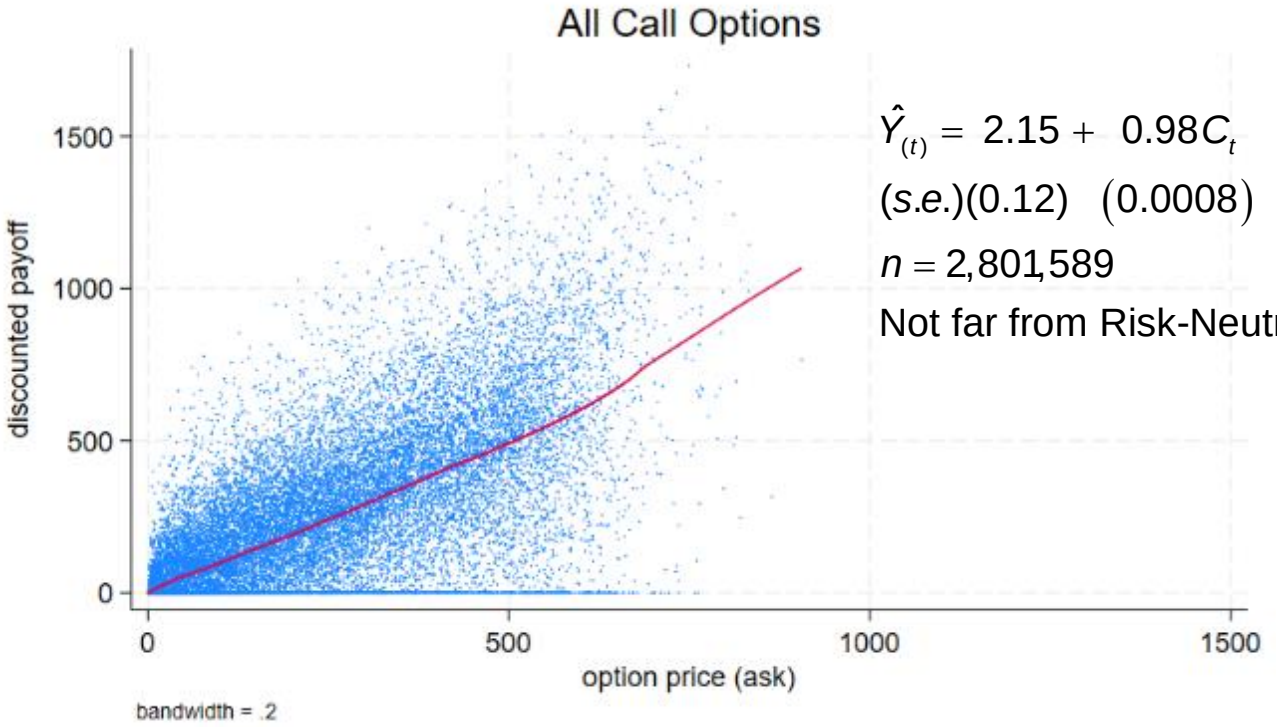


DATA

- All European Calls and European Puts written on S&P500 Index
  - Settlement between 3 Jan 2022 and 29 December 2023
    - Bear Market: 3 Jan 2022 – 20 June 2022
    - "Sideways Market": 21 June 2022 – 12 Oct 2022
    - Bull Market: 13 Oct 2022 – 29 Dec 2023
- Moneyness between 0.85 and 1.15
    - 2,801,589 Call Options
    - 2,912,356 Put Options



RESULTS



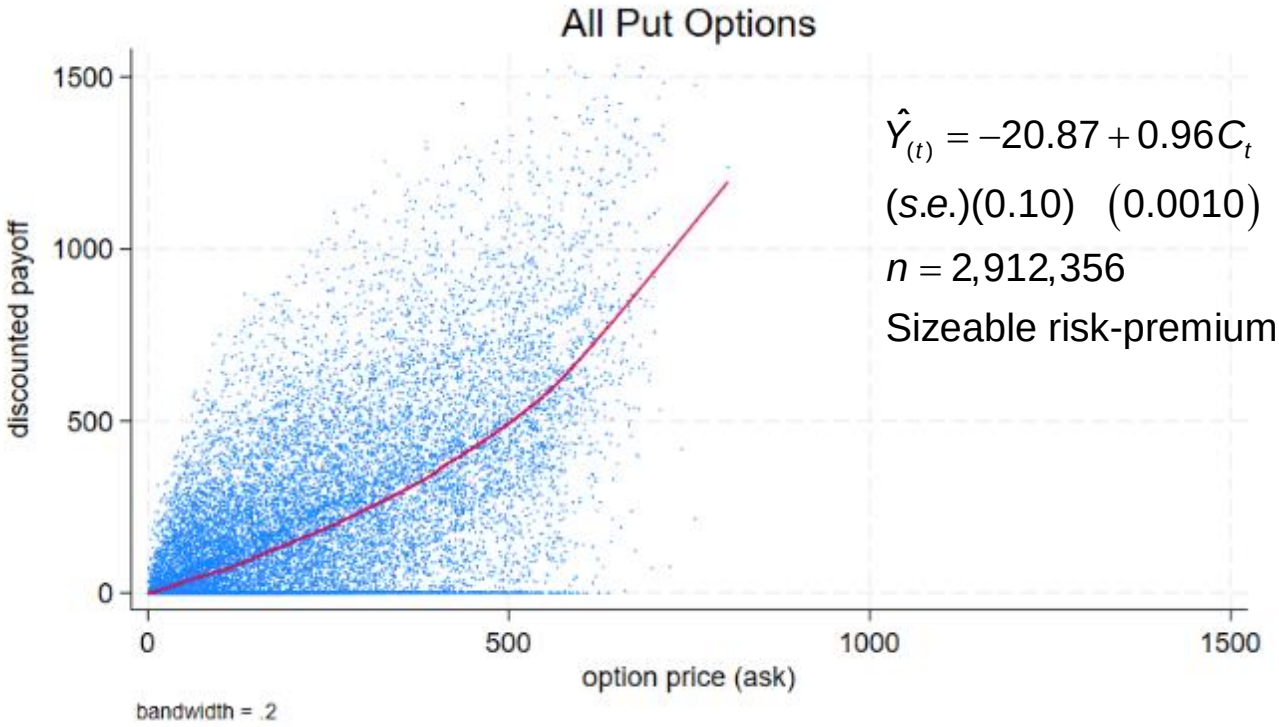
$$\hat{Y}_{(t)} = 2.15 + 0.98C_t$$

$$(s.e.)(0.12) \quad (0.0008)$$

$$n = 2{,}801{,}589$$

Not far from Risk-Neutral Prediction

RESULTS



$$\hat{Y}_{(t)} = -20.87 + 0.96 C_t$$

$$(s.e.)(0.10) \quad (0.0010)$$

$$n = 2{,}912{,}356$$

Sizeable risk-premium: 20.87

Results Separately by Market-type

CALLS	β_0	β_1	Risk Premium?
Bear	-6.41	0.44	Yes
Bull	4.60	1.27	Negative
Sideways	-21.60	0.90	Yes
PUTS			
Bear	5.40	1.30	Negative
Bull	-13.65	0.59	Yes
Sideways	-7.10	0.81	Yes

Summary

- Framework allows direct estimation of Risk Premium from large amounts of data.
  - Direct estimation, unlike e.g. Variance Risk Premium (VRP) approach
- Results from full sample:
  - Call market very close to RN prediction
    - Sellers of Calls do NOT need to be compensated for risk of "bullish shocks"
  - Put market displays high Risk Premium
    - Sellers of Puts DO need to be compensated for risk of occasional large losses during market crashes.

END